

PART B Questions -2

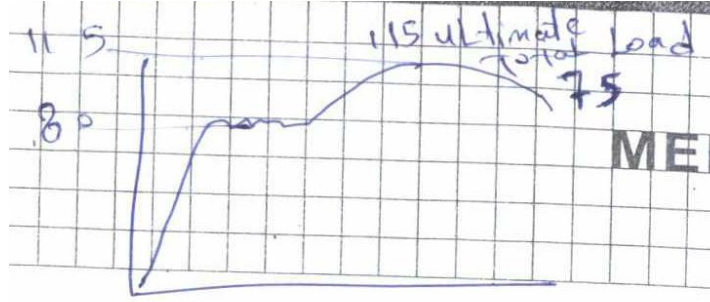
- 1 Examine the partially completed P2-3 weld on the pipe between marker 12 and marker 1 on an 8.7 mm thickness. What is the approximate weld deposit thickness: ⁽¹⁾
(A) 18 mm
(B) 8.7 mm
(C) 6.2 mm
(D) 5.5 mm
(E) 3.5
- 2 Examine T joints form EF1 to EF6. Which one has visible craters? ⁽²⁾
(A) EF2
(B) EF3
(C) EF4
(D) EF5
(E) EF6
- 3 Examine EF6 where its welding symbol on the DWG shown. Which one best describes it: ⁽³⁾

(A) accepted
(B) reject – due to undersize, undercut
(C) reject – due to undersize, overlap
(D) reject – due to undersize only
(E) reject – due to overlap only
- 4 Examine the P1-0 end of the pipe. What is the pipe schedule at that end: ⁽⁴⁾
(A) 40
(B) 80
(C) 120

- (D) 160
(E) STD
- 5 Examine bend B2 preparation: ⁽⁵⁾
(A) accepted
(B) unaccepted – due to width
(C) unaccepted – edge radius
(D) unaccepted – grinding direction
(E) unaccepted – reinforcement
- 6 Examine the groove weld EG2-3 reinforcement of steel structure, assume dissimilar thickness of 50 and 65 mm: ⁽⁶⁾
(A) reject – > 2 mm
(B) reject – > 3 mm
(C) reject – > 5 mm
(D) accept – ≤ 3 mm
(E) accept – ≤ 5 mm
- 7 Examine EF2 T joint of steel structure with static loading: ⁽⁷⁾
(A) accepted
(B) rejected – overlap
(C) rejected – porosity
(D) rejected – UC
(E) rejected – unequal leg size
- 8 Examine the middle weld on pressure piping between marker 5 and 10, where the thickness is 8.7 mm, design temp. 150°C, the weld quality is: ⁽⁸⁾
(A) accepted
(B) rejected – IF
(C) Rejected – spatter
(D) rejected – excess reinforcement
(E) both B and C

- 9 Examine the upstream pipe of pressure piping, plus VT which tests of inspection shall be done for the finished surface after repair welding is completed of the excavation in this weld? ⁽⁹⁾
- (A) no additional tests
 - (B) PT
 - (C) MT
 - (D) UT
 - (E) B or C
- 10 For pipeline, which inspection is required prior to welding the repair excavation on the upstream weld? ⁽¹⁰⁾
- (A) VT
 - (B) PT or MT
 - (C) RT
 - (D) All above
 - (E) both A and B
- 11 Examine the down stream partially welded pipe between marker 2 and 3 of pressure piping, which one is correct? ⁽¹¹⁾
- (A) tack weld must be removed لو فيه شروخ
 - (B) tack weld may be incorporated in weld
 - (C) tack weld end shall be ground before proceeding لو مفيش شروخ
 - (D) tack weld is irrelevant if tye other tacks are sound
 - (E) both B and C
- 12 Examine the internal alignment of partially welded pipe at marker 4 of pressure piping. Which one is true? ⁽¹²⁾
- (A) rejected - > 2 mm
 - (B) rejected - > 0.8 mm and < 2 mm
 - (C) rejected - > 2.5 mm and < 5 mm
 - (D) accepted - > 2.5 mm and < 3 mm
 - (E) accepted - < 2 mm

- 13 During a tensile test, it hesitated at 80 KN, then rise to a max load of 115 KN, then load dropped to 75 KN, then failed. What information must be recorded in space 57 of Annex VIII? ⁽¹³⁾



- (A) 75 KN
 (B) 80 KN
 (C) 115 KN
 (D) 155 KN
 (E) 275 KN
- 14 Examine exhibit PQR No. 002. A WPS based on this PQR would be qualified for what base metal thickness range? ⁽¹⁴⁾
- (A) 5 mm – 50 mm
 (B) 13 mm – 50 mm
 (C) 25 mm – 33.25 mm
 (D) 5 mm – 200 mm
 (E) 3 mm – unlimited
- 15 The initial specimen dimension of a rectangular tensile bar were 20 mm by 13 mm. What information should be recorded in space 56 of annex VIII? ⁽¹⁵⁾
- (A) 26
 (B) 33
 (C) 52
 (D) 260
 (E) 520

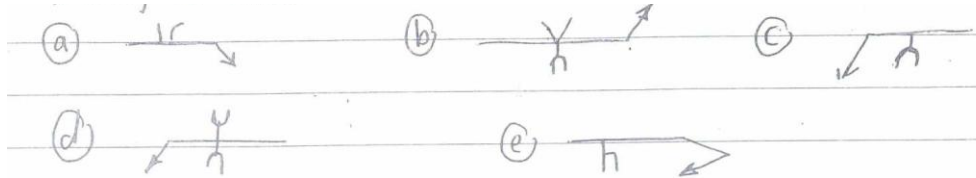
- 16 Examine the tensile test results on PQR No. 001. Annex III provides the following:
Material A106 Gr C Min Tensile: 485 MPa
which of the following is true? ⁽¹⁶⁾
- (A) accepted
 - (B) rejected – weld metal failure is below the base metal SMTS
 - (C) rejected – both values below the base metal SMTS
 - (D) rejected – one value more than 5% below metal SMTS
 - (E) rejected – both values more than 5% below metal SMTS
- 17 Examine exhibit PQR No. 001 and Annex I of the BOS, the weld metal deposit chemistry of the E4918 (E7018) electrode is:
C: 0.04% Cr: 0.05% Mo: 0.0%
Ni: 0.06% Mn: 0.60% Si: 0.41%
which one is true regarding A number recorded in the PQR? ⁽¹⁷⁾
- (A) it is correct
 - (B) incorrect should be 5
 - (C) incorrect should be 4
 - (D) incorrect should be 3
 - (E) incorrect should be 2
- 18 Examine the bend specimens. Which one is not ground as recommended? ⁽¹⁸⁾
- (A) B2
 - (B) B3
 - (C) B4
 - (D) B5
 - (E) B6
- 19 Examine bend B1, disregard any discontinuities, which one is true regarding preparation and bending? ⁽¹⁹⁾
- (A) accepted
 - (B) rejected – weld not within bend portion
 - (C) rejected – edge radius

- (D) rejected – grinding direction
(E) both B and D
- 20 Examine PWHT chart of exhibit # 6, what value must be entered on the line 38 of PQR shown in Annex VIII? ⁽²⁰⁾
- (A) 1350 F
(B) 1400 F
(C) 1450 F
(D) 3 hours
(E) 5 hours
- 21 Examine Figure 1, the plastic bottle contains: ⁽²¹⁾
- (A) cleaning solvent
(B) rust preventative
(C) PT
(D) high viscosity lubricant
(E) UT couplant
- 22 Examine exhibit PQR No. 001 for steel base material.
For a welder qualification record WQTR based on this PQR:
The welder is not qualified to weld which one? ⁽²²⁾
- (A) M1 Gr 2
(B) M11B
(C) M21
(D) M4
(E) M5C
- 23 Examine B2 for pressure piping. Which one is correct disposition of this B2? ⁽²³⁾
- (A) disregarded – incorrect grinding
(B) disregarded – improper edge radius
(C) disregarded – weld is not centered

- (D) reject – defect > 3 mm
(E) reject – open defect > 6 mm
- 24 Examine the exhibit WQTR – B, for which material the welder is qualified? ⁽²⁴⁾
(A) A36
(B) A106 Gr C
(C) API 5L X80
(D) All the above
(E) Only A or B
- 25 Examine the Exhibit WQRT-B. Is the welder qualified to weld the following: ⁽²⁵⁾
Material: A36 Progression: Vertical uphill
Backing: metal backing
(A) Yes
(B) No, for A36
(C) No, for vertical up
(D) No, for backing
(E) Both B & C
- 26 Examine the Exhibit WQRT-C. Is the welder qualified to weld the following: ⁽²⁶⁾
Electrode: E4310 (E6010) Progression: Downhill
Deposit: 5 mm thickness
(A) Yes
(B) No, for E4310
(C) No, for downhill
(D) No, for deposit thickness
(E) Both B & C

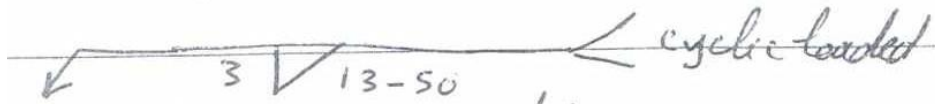
- 27 Examine the Exhibit WQRT-E and the pipe specimens. Which weld is this welder qualified using only the SMAW, W.T. = 7 mm? ⁽²⁷⁾
- (A) upstream weld
 - (B) middle weld
 - (C) downstream weld
 - (D) none above
 - (E) only A and C

- 28 Examine the downstream pipe edge preparation, assume a pipe with an identical edge preparation is fitted to this edge. Which of the following is correct? ⁽²⁸⁾



- 29 Examine the upstream pipe end edge preparation bevel angle on the pipe specimen for compliance with the the exhibit WPS GMAW -3. Which one is true regarding the edge preparation? ⁽²⁹⁾
- (A) reject – single bevel edge shape not permitted by the WPS
 - (B) reject – bevel angle > permitted by the WPS
 - (C) reject – bevel angle < permitted by the WPS
 - (D) both B and C – bevel angle varies between markers 7 and 8
 - (E) accepted bevel angle per WPS
- 30 Examine the root opening of the pipe downstream weld at marker 1-6 for compliance with exhibit 8 WPS 6. What is describing the root opening? ⁽³⁰⁾
- (A) reject – due to > 5 mm
 - (B) reject – due to < 2.5 mm
 - (C) reject – due to > 2.5
 - (D) reject – due to > 2 mm and < 2.5 mm
 - (E) accepted

- 31 Examine Exhibit Figure 2. What is performing: ⁽³¹⁾
- (A) scrubbing contamination off the transducer
 - (B) angle beam calibration check
 - (C) straight beam calibration check
 - (D) getting the sensitivity level
 - (E) demagnetizing the transducer
- 32 Examine EE-3 thermal cut edge roughness of steel structure for mechanized FCAW. Which one best describes it? ⁽³²⁾
- (A) Accepted
 - (B) Unaccepted without additional preparation
 - (C) Unaccepted because a machined edge is required
 - (D) Unaccepted because of mill induced discontinuity
 - (E) both B and D
- 33 Examine lap joints at corners A and D of steel structure according to the welding symbol shown:



- Which one is true regarding starts and terminations at A and D? ⁽³³⁾
- (A) unaccepted – no wrap around at A
 - (B) unaccepted – no wrap around at D
 - (C) unaccepted – either A or D
 - (D) unaccepted – no hold back at L4
 - (E) accepted
- 34 Examine groove EG2-5 of pipeline criteria. Which one is true? ⁽³⁴⁾
- (A) rejected – unrepaired shallow crack
 - (B) rejected – cluster porosity
 - (C) rejected – arc burn
 - (D) rejected – both B and C
 - (E) accepted

- 35 Examine groove EG2-4 of pipeline criteria. Which one is true, if the fitted groove width is 25 mm? ⁽³⁵⁾
- (A) rejected – lack of fusion
 - (B) rejected – under cut (UC)
 - (C) rejected – crown height is smaller than the specified.
 - (D) rejected – the weld face is wider than the specified
 - (E) both B and D
- 36 Examine midweld P2-2 between marks 4 and 6 of pipeline. Weld quality is? ⁽³⁶⁾
- (A) accepted
 - (B) rejected – EU depth
 - (C) rejected – CP diameter
 - (D) rejected – IF length
 - (E) both B and C
- 37 Examine upstream of pipeline at markers 1-5, based on thickness 8.7 mm. The weld quality is? ⁽³⁷⁾
- (A) accepted
 - (B) rejected – EU depth
 - (C) rejected – beads start and terminate at same location
 - (D) rejected – spatter
 - (E) both C and D

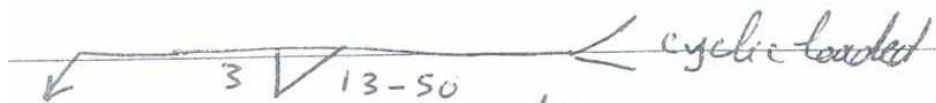
dimension
 (16) examine downstream weld of pipe welded with SMAW using
 WPS GTAW/SMAW-2A for pipeline where thickness is 8.7 mm
 the weld quality is?
 (a) Rejected – EU depth
 (b) Rejected – excessive crown
 (c) " – spatter
 (d) all above
 (e) only (A) and (C)

- 38 Examine downstream weld of pipe welded with SMAW using WPS SMAW-1 for pipeline, where thickness is 8.7 mm. The weld quality is? ⁽³⁸⁾

(A) rejected – EU depth

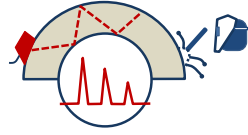
- (B) rejected – excessive crown
(C) rejected – spatter
(D) all above
(E) only B and C
- 39 Examine the partially completed downstream weld of pressure piping at markers 4-5, where thickness is 8.7 mm. Is there sufficient weld deposit to allow interruption of the 95°C preheat and welding?⁽³⁹⁾
- (A) yes
(B) no – filling < 25%
(C) no – filling < 50%
(D) no – filling < 75%
(E) no – groove must be completely filled
- 40 Examine exhibit WPS FCAW-2. During an in process inspection, the inspector noted that:
- welding position: flat, horizontal, vertical
 - interpass temperature : 120°C
 - wire speed: 4.8 m/min
 - shielding gas: 75% Ar + 25% CO₂ at 21 l/min
 - electrode dia.: 1.2 mm
 - electrode type: E490T-1M (E70T-1M)
- Is the welder in compliance with this WPS?⁽⁴⁰⁾
- (A) Yes
(B) no – interpass temperature
(C) no – wire speed
(D) no – welding position
(E) both B and D
- 41 Examine the tensile test calculations for the ultimate unit stress on PQR No. 007B. What are the correct ultimate unit stress values for the specimens 1 and 2?⁽⁴¹⁾
- (A) specimen 1: 16,056 lbs specimen 2: 14,939 lbs

- (B) specimen 1: 103,042 psi specimen 2: 94,279 psi
 (C) specimen 1: 102,924 psi specimen 2: 94,551 psi
 (D) specimen 1: 103,238 psi specimen 2: 94,077 psi
 (E) specimen 1: 104,088 psi specimen 2: 95,212 psi
- 42 Examine exhibit PQR No. 002. Which WPS variable is not qualified by the PQR for: ⁽⁴²⁾
 process: FCAW material: A369 Gr FP1
 pipe size: 200 mm pipe thickness: XXS
 weld: V-groove with backing position: flat
 filler: E491T (E71T) preheat/interpass: 95°C/260°C
 PWHT: N.A. shielded gas: 75%Ar + 25%CO₂
 (A) not qualified – position and material
 (B) not qualified – material and filler metal
 (C) not qualified – gas composition and flow rate
 (D) not qualified – filler metal and preheat/interpass temperature
 (E) not qualified – material and preheat/interpass temperature
- 43 Examine exhibit WPS P91-4 and P91 heat treatment chart. Is the holding time comply with WPS for 100 mm thickness? ⁽⁴³⁾
 (A) Yes
 (B) No – holding time < WPS
 (C) No – holding time > WPS
 (D) No – WPS does not permit PWHT
 (E) No – it is < 15 minutes
- 44 Examine lap joints at corners B and C of steel structure according to the welding symbol shown:



Which one is true regarding starts and terminations at B and C? ⁽⁴⁴⁾

- (A) unacceptable – no wrap around at C
 (B) accepted – around B



- (C) unaccepted – either B or C
- (D) unaccepted – no hold back at edge C
- (E) both (A) and (B)



Figure 1



Figure 2

Answers

- 1 - C
- 2 - B
- 3 - A
- 4 - B
- 5 - B
- 6 - E
- 7 - D
- 8 - B (IF is the cause of rejection, spatter is acceptable since it is not on the BM)
- 9 - E
- 10 - E
- 11 - E
- 12 - E
- 13 - C
- 14 - D
- 15 - D
- 16 - B
- 17 - A
- 18 - A
- 19 - C
- 20 - B
- 21 - E
- 22 - C
- 23 - D
- 24 - D
- 25 - C
- 26 - E
- 27 - D
- 28 - C
- 29 - C
- 30 - C
- 31 - C
- 32 - B
- 33 - B
- 34 - A
- 35 - D
- 36 - A (unacceptable if IF > 25 mm, or > 22 mm in the pipe of 88 mm diameter)
- 37 - E
- 38 - E
- 39 - A
- 40 - A
- 41 - C

- 42 - E
43 - C
44 - E